

Cabe Robertson

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OBJECTIVE

Pursuing Internship/Co-op roles for Summer/Fall 2026

EDUCATION

New Mexico State University - Las Cruces, NM

Expected December 2027

- Bachelor of Science in Electrical Engineering
(Communications and Signal Processing Concentration)

SKILLS

SOFTWARE: Arduino IDE, C, MATLAB, EAGLE, VHDL, Python, Overleaf

HARDWARE: Oscilloscope, Power Supply, Anemometer, Function Generator, FPGA, Soldering

LANGUAGE: English (Native), Spanish (Native)

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern, GILZ LLC - Remote

Mar 2026 - May 2026

- Deconstructed a desk blower fan to test the internal BLDC motor's voltage/current ratings and PWM capabilities for use in an air filtration system
- Drafted an Arduino program on an ESP32 to control the 3 speeds of the BLDC motor to add user control capabilities
- Conducted benchmarks with the original blower fan to measure noise, temperature, speed, and velocity of the fan using different attachments such as air ducts and filters

Hardware Engineering Intern, Syndetix Incorporated - Las Cruces, NM

Apr 2025 - Aug 2025

- Created an affordable demonstration kit for \$219 to accurately present the capabilities of a communication disrupting device to law enforcement and defense agencies while also being easily produceable
- Improvised a lightweight solution for a scenario in which wind could knock over the demonstration, ensuring portability and stability of the kit
- Formulated an in-depth user manual for a new product, guiding customers on proper handling and set up of the device for missions or training
- Repaired a critical permission-based synchronization issue on the company server that had been active for 4 years, improving efficiency of workstations company wide

Research Assistant, Mitchell Coding Group - Las Cruces, NM

Aug 2025 - Feb 2026

- Developed Python programs to simulate channel decoding using Linear Algebra for wireless communication models
- Executed experiments with multiple hour runtimes utilizing contrasting bit string values to ensure software functionality
- Wrote MATLAB scripts to process and visualize experimental results, producing publication-quality plots and diagrams

ACADEMIC PROJECTS

Self-Navigating Robot, (Arduino)

- Developed a fully functional program in Arduino IDE that correctly avoids obstacles autonomously using ultrasonic sensors
- Validated the functionality of the program by testing maneuverability when facing hurdles of varying directions
- Refined code over many trials to maximize the programs efficiency in object avoidance

Seven-Segment Display, (FPGA, VHDL)

- Programmed a Seven-Segment Display in a VHDL that correctly displayed numbers and characters on an FPGA board
- Orchestrated trials of the program by testing all possible BCD (0-9) and HEX (A-F) inputs via console GUI
- Simplified Boolean expressions using truth tables and Karnaugh maps for code optimization

LEADERSHIP

Vice President, IEEE NMSU Chapter

Oct 2024 - Present

- Directing a unique fundraising project with an estimated revenue of \$50,000 to revitalize the chapter and ECE department
- Oversee and manage all club projects ensuring tasks are being completed efficiently, and groups are receiving adequate support
- Mentor students during technical workshops, teaching proper troubleshooting methods and soldering techniques
- Diplomat for communication with industry representatives, external IEEE chapters, and other student organizations on campus for sponsorship proposals and collaborations